PATH CONNECTEDNESS AND HOLOMORPHIC-CURVE CONFINEMENT FOR NORMALIZED COMPLETE ALMOST COMPLEX STRUCTURES

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Allowed standard prerequisites (editorial).

- Symplectic cutting, holomorphic-curve compactness and standard toric/Fukaya background as used by author. Original Sikorav, STVFluxShape and FOOO/ChoOh bibliography retained; actual source-local arguments included.

Source reference. Original reference 1.22: algebra-geometry/sources/sigma-2026-066/native/sigma26-066.tex, source lines 381–381. Later application or generalization mentioned in the narrative; not an input to the retained target proof.

0.1. **The results.** As our first proxy for completeness, we use the well-established notion of *geometric boundedness* (see, e.g., [14, Section 4.2]): an ω-compatible almost complex structure J on M is called *geometrically bounded* if the associated Riemannian metric $g_J(\cdot,\cdot) = \omega(\cdot, J\cdot)$ is complete and has uniformly bounded sectional curvature and uniformly positive injectivity radius.[1]

We begin with an existence question.

Question 0.1. Given ω, does there exist a geometrically bounded ω-compatible almost complex structure J on M?

It is customary to call (M,ω) itself geometrically bounded if the answer to Question 0.1 is affirmative. We consider this question when we start with a geometrically bounded M and excise a subset $N \subset M$. Question 0.1 can be understood as the question: *can the distance to N of any compact subset of $M \setminus N$ be made arbitrarily large by adjusting the almost complex structure?*

Theorem 0.2. *Let M be a geometrically bounded symplectic manifold with finitely generated spherical homology, and suppose ω is rational. Let $N \subset M$ be a properly embedded symplectic hypersurface. Then $M \setminus N$ is not geometrically bounded.*

0.2. **Completeness in symplectic topology.** The obstructions to geometric boundedness in Question 0.1 come from J-holomorphic curves with finite energy connecting distant compact subsets. This suggests viewing geometric boundedness as a symplectic counterpart of geodesic completeness [11], with J-holomorphic curves playing the role of geodesics.

However, geometric boundedness has drawbacks: it involves curvature (second derivatives), and the space of geometrically bounded structures may not be contractible or even connected. At the other extreme, one can consider the weakest completeness property.

Definition 0.3 (J-completeness)**.** An ω-tame almost complex structure J on a symplectic manifold (M,ω) is said to be *J-complete* if the associated Riemannian metric g_J is complete and for every compact subset $K \subset M$ and every $E > 0$ there exists

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[1] To treat ω-tame almost complex structures, one must also impose a uniform bound on the norm of ω with respect to g_J (i.e., $|\omega|_{g_J} \leq C$).

$R > 0$ such that any connected J-holomorphic curve whose boundary meets both K and $M \setminus B_R(K)$ has energy at least E.

This is stronger than metric completeness but too weak to control the topology of the space of J-complete structures.

We now consider an intermediate notion between J-completeness and geometric boundedness which is of a C^0 nature and which gives rise to a contractible condition. It is closely related to that of intermittent boundedness which was introduced in [6]. For each J, we construct in Section 2 an intrinsically defined continuous *scale function* $\rho_J \colon M \to (0, +\infty]$ measuring the isoperimetric scale at each point. The *normalized distance* $d_{\mathrm{norm},J}$ measures path lengths in units of ρ_J

$$L_{\mathrm{norm},J}(\gamma) := \int_0^1 \rho_J(\gamma(t))|\dot{\gamma}(t)|_{g_J} \mathrm{d}t,$$

with $d_{\mathrm{norm},J}$ the induced path metric. The scale function depends on the choice once and for all of a cutoff constant $c_0 > \frac{1}{4\pi}$. The constant c_0 measures the allowable deviation of the metric from being flat whereas the scale function at a point p measures the size of a neighborhood in which the metric does not deviate from flatness more than is allowed by the cutoff constant.

Definition 0.4 (normalized completeness)**.** An ω-compatible almost complex structure J is *normalized complete* if the metric space $(M, d_{\mathrm{norm},J})$ is complete.

Remark 0.5. In this definition and Theorem 0.6, we focus on the ω-compatible case. The statements are true for the ω-tame case with a very slight adjustment. See Remark 2.10.

Theorem 0.6 (basic properties of normalized completeness)**.** *Let (M, ω) be a symplectic manifold.*

(1) Connectedness. *The space of normalized complete ω-compatible almost complex structures is path connected.*
(2) Confinement and J-completeness. *If J is normalized complete, then J is J-complete (Definition 0.3). More precisely, any connected J-holomorphic curve $u \colon (\Sigma, \partial\Sigma) \to (M, \omega)$ satisfies a confinement estimate of the form*

$$u(\Sigma) \subset B^{d_{\mathrm{norm},J}}_{CE(u)+C'}(u(\partial\Sigma)),$$

where $E(u) = \int_\Sigma u^*\omega$ and the constants $C, C' > 0$ are universal constants.
(3) Equivalence to standard metric under uniform bounds. *If g_J has uniformly bounded sectional curvature and uniformly positive injectivity radius, then $d_{\mathrm{norm},J}$ and d_{g_J} are bi-Lipschitz equivalent. In this case, normalized completeness is equivalent to metric completeness of g_J.*

1. Proofs

1.1. **Energy confinement for J-holomorphic curves.** Throughout, if $u \colon (\Sigma, \partial\Sigma) \to (M, \omega)$ is a J-holomorphic curve with (possibly empty) boundary, its energy is $E(u) := \int_\Sigma u^*\omega \geq 0$.

The following confinement lemmas are stated for ω-compatible J. At the end of this subsection we note that both hold for ω-tame J under the uniform tameness condition from the footnote to the definition of geometric boundedness on page 1 (in Section 1.1).

Lemma 1.1 (Sikorav's confinement estimate)**.** *Let (M, ω) be a symplectic manifold and let J be an ω-compatible almost complex structure such that the associated metric $g_J(\cdot, \cdot) := \omega(\cdot, J\cdot)$ is complete and has uniformly bounded sectional curvature and positive injectivity radius (i.e., J is geometrically bounded). Then there exist*

a constant $r_0 > 0$ depending on the geometry and a universal constant $c > 0$ such that for every connected J-holomorphic curve u one has

$$u(\Sigma) \subset B_{r_0 \lceil E(u)/(cr_0^2) \rceil}(u(\partial\Sigma)),$$

where $B_r(\cdot)$ denotes the closed g_J-ball of radius r about a subset and $\lceil \cdot \rceil$ is the ceiling function.

Proof. See [14, Proposition 4.4.1]. □

This confinement estimate is concerned with the large scale. Namely, it contains no information about curves with energy much smaller than cr_0^2 other than that the diameter is $\leq r_0$. In some cases, e.g., in the proof of Proposition 1.22, we need an estimate for the small scale. This is taken care of by the following lemma.

Lemma 1.2 (improved confinement)**.** *In the setting of Lemma* 1.1*, with the same r_0 and c, and letting $C := 1/c$, one has*

$$u(\Sigma) \subset B_\rho(u(\partial\Sigma)), \qquad \rho := \max\bigl(3CE(u)/r_0, \sqrt{CE(u)}\bigr).$$

Proof. The argument relies on the same monotonicity inequality used for Sikorav's lemma: if a J-holomorphic curve meets the center of a ball of radius $r \leq r_0$ and the boundary of the J-holomorphic curve does not meet the interior of the ball, then the energy in that ball is at least cr^2.

If $E(u)/\bigl(cr_0^2\bigr) \leq 1$, we apply the monotonicity inequality directly: if the image contained a point at the center of a ball of radius greater than $\sqrt{E(u)/c}$ whose interior did not meet $u(\partial\Sigma)$, then monotonicity would force $E(u) \geq cr^2 > E(u)$ (since $E(u) \geq cr^2$ implies $r \leq \sqrt{E(u)/c}$). Hence $u(\Sigma)$ is contained in a ball of radius $\sqrt{E(u)/c} = \sqrt{CE(u)}$ about $u(\partial\Sigma)$.

If $E(u)/\bigl(cr_0^2\bigr) > 1$, take the maximum number n of disjoint balls of radius r_0 that the image of u can meet. Then $n \leq E(u)/\bigl(cr_0^2\bigr)$ by monotonicity, and the image is contained in a ball of radius $(2n+1)r_0 \leq r_0\bigl(1 + 2E(u)/\bigl(cr_0^2\bigr)\bigr) < 3E(u)/(cr_0) = 3CE(u)/r_0$ around $u(\partial\Sigma)$. □

For later use, we also record a localized version, originally observed in [6].

Definition 1.3. For $a > 0$, we say that the complete Riemannian metric g_J is *a-bounded at a point $p \in M$* if $\mathrm{inj}_{g_J}(p) \geq \frac{1}{a}$ and $\sec_{g_J} \leq a^2$ on $B_{1/a}(p)$. We say that g_J is *a-bounded on a set* if it is a-bounded at every point of that set.

Remark 1.4. When g_J is a-bounded, we can take $r_0 = 1/(2a)$ [12, Theorem 4.4.1 and Remark 4.4.3].

Lemma 1.5 (local confinement)**.** *Fix $a > 0$. There exist $r_0 = r_0(a) > 0$ and a universal constant $C > 0$ with the following property. Let $K \subset M$ be compact and let $R > 0$. Assume J is ω-compatible and g_J is complete and is a-bounded on the g_J-ball $B_R(K)$. Let u be a J-holomorphic curve u with $u(\partial\Sigma) \subset K$ and write*

$$\rho := \max\bigl(2CE(u)/r_0, \sqrt{CE(u)}\bigr).$$

If $\rho \leq R$, then $u(\Sigma) \subset B_R(K)$.

Proof. The proof of Lemma 1.2 relies on the monotonicity inequality which depends only on the local geometry of a complete Riemannian metric. So taking r_0 and C as in Lemma 1.2, we obtain by the same argument that $u(\Sigma) \subset B_\rho(u(\partial\Sigma))$. □

2. Normalized completeness

2.1. **An isoperimetric scale.** Let (M,ω) be a symplectic manifold and let J be an ω-compatible almost complex structure. Write $g_J(\cdot,\cdot) := \omega(\cdot, J\cdot)$ for the associated Riemannian metric, and write $\ell_{g_J}(\gamma)$ for the g_J-length of a piecewise smooth loop $\gamma \subset M$.

Definition 2.1 (isoperimetric ratio on a ball)**.** For $p \in M$ and $r > 0$, define

$$\mathcal{I}_J(p;r) := \sup_{\gamma} \inf_{u} \frac{\int_D u^*\omega}{\ell_{g_J}(\gamma)^2} \in [0,+\infty],$$

where the supremum ranges over all *contractible* piecewise smooth loops $\gamma \subset B_r^{g_J}(p)$ with $\ell_{g_J}(\gamma) > 0$, and the infimum ranges over all *smooth* discs $u\colon (D,\partial D) \to \big(B_r^{g_J}(p), \gamma\big)$ with $\partial u = \gamma$.

Fix once and for all a cutoff constant $c_0 > 0$.

Definition 2.2. For $p \in M$, let

$$r_{\mathrm{iso},J}(p) := \sup\{r \in [0,+\infty) \mid \mathcal{I}_J(p;r) \le c_0\}$$

and define the *isoperimetric scale* $\rho_J(p) := \min\{1, r_{\mathrm{iso},J}(p)\} \in [0,1]$.

Remark 2.3. The constant c_0 is an arbitrary fixed numerical cutoff. The effect of changing c_0 is discussed below.

Lemma 2.4 (properties of the scale function)**.** *The function $\rho_J\colon M \to [0,1]$ is positive everywhere and 1-Lipschitz with respect to d_{g_J}. In particular, ρ_J is bounded away from 0 on compact subsets.*

Proof. Positivity follows from the fact that inside a small enough geodesic ball one has curvature bounds, hence a quadratic isoperimetric inequality for filling contractible loops by discs; see [12, Theorem 4.4.1].

Fix $p, q \in M$ and set $\delta := d_{g_J}(p,q)$. Then for every $r > \delta$ one has the inclusion $B^{g_J}_{r-\delta}(q) \subset B^{g_J}_r(p)$. By Definition 2.1, this implies

$$\mathcal{I}_J(q; r-\delta) \le \mathcal{I}_J(p;r).$$

Consequently, if $\mathcal{I}_J(p;r) \le c_0$, then $\mathcal{I}_J(q;r-\delta) \le c_0$, hence $r_{\mathrm{iso},J}(q) \ge r_{\mathrm{iso},J}(p) - \delta$. By symmetry we also obtain $r_{\mathrm{iso},J}(p) \ge r_{\mathrm{iso},J}(q) - \delta$, so $|r_{\mathrm{iso},J}(p) - r_{\mathrm{iso},J}(q)| \le d_{g_J}(p,q)$. This proves the claim, and hence the same for $\rho_J = \min\{1, r_{\mathrm{iso},J}\}$. □

2.2. **Normalized length and normalized completeness.**

Definition 2.5 (normalized length and distance)**.** Let J be ω-compatible and let ρ_J be as in Definition 2.2. For a piecewise smooth path $\gamma\colon [0,1] \to M$, define its *normalized length* by

$$L_{\mathrm{norm},J}(\gamma) := \int_0^1 \rho_J(\gamma(t)) |\dot\gamma(t)|_{g_J} \mathrm{d}t.$$

The induced path metric is denoted $d_{\mathrm{norm},J}$. Equivalently, $d_{\mathrm{norm},J}$ is the distance associated to the conformal metric $\widetilde{g}_J := \rho_J^2 g_J$.

Remark 2.6. Note that the metric $\widetilde{g}_J$ is generally not smooth.

Definition 2.7 (normalized completeness)**.** An ω-compatible almost complex structure J is *normalized complete* if the metric space $(M, d_{\mathrm{norm},J})$ is complete.

Lemma 2.8 (closed bounded balls are compact)**.** *If J is normalized complete, then for any basepoint $p_0 \in M$ and any $n \in \mathbb{N}$, the closed $d_{\mathrm{norm},J}$-ball $K_n := \overline{B}_n^{d_{\mathrm{norm},J}}(p_0)$ is compact in the manifold topology.*

Proof. Since ρ_J is continuous and positive, we can approximate it by a smooth positive function $\widetilde{\rho}_J$ such that the ratio $\rho_J/\widetilde{\rho}_J$ is uniformly bounded away from 0 and $+\infty$. The smooth Riemannian metric $\widehat{g}_J := \widetilde{\rho}_J^2 g_J$ is then uniformly equivalent to $\widetilde{g}_J = \rho_J^2 g_J$, hence the distances $d_{\widehat{g}_J}$ and $d_{\mathrm{norm},J}$ are bi-Lipschitz equivalent. Since $(M, d_{\mathrm{norm},J})$ is complete, $(M, d_{\widehat{g}_J})$ is also complete. Each closed ball $K_n = \overline{B}_n^{d_{\mathrm{norm},J}}(p_0)$ is bounded with respect to $d_{\widehat{g}_J}$ (by uniform equivalence), and since $\widehat{g}_J$ is smooth, the Hopf–Rinow theorem implies that closed bounded sets are compact. Thus each K_n is compact in the manifold topology. □

2.3. **Proof of Theorem 0.6.**

2.3.1. *Connectedness (Theorem 0.6 (1)).*

Proof. The proof of connectedness is precisely the zigzag construction in [6, Theorem 4.7].

Denote by $\mathcal{J}_\omega^{\mathrm{comp}}$ the space of smooth ω-compatible almost complex structures with the C^∞ topology and by $\mathcal{J}_\omega^{\mathrm{nc,comp}} \subset \mathcal{J}_\omega^{\mathrm{comp}}$ the subspace of normalized complete structures. Assume $\mathcal{J}_\omega^{\mathrm{nc,comp}} \neq \varnothing$. Without loss of generality, assume that M is connected and fix a basepoint $p_0 \in M$. Given $J_0, J_1 \in \mathcal{J}_\omega^{\mathrm{nc,comp}}$, consider the closed normalized balls

$$B_n^0 := \overline{B}_n^{d_{\mathrm{norm},J_0}}(p_0), \qquad B_n^1 := \overline{B}_n^{d_{\mathrm{norm},J_1}}(p_0), \qquad n \in \mathbb{N}.$$

By Lemma 2.8, each of these balls is compact. By construction of the normalized distance, successive balls in each sequence are at normalized distance at least 1

$$d_{\mathrm{norm},J_0}\big(B_n^0, M \setminus B_{n+1}^0\big) \geq 1, \qquad d_{\mathrm{norm},J_1}\big(B_n^1, M \setminus B_{n+1}^1\big) \geq 1, \qquad n \in \mathbb{N}. \tag{2.1}$$

We choose inductively increasing subsequences $n_1^0 < n_2^0 < \cdots$ and $n_1^1 < n_2^1 < \cdots$ in $\mathbb{N}$ with the following nesting property: for each $i \geq 1$,

$$B_{n_i^0+1}^0 \subset \operatorname{Int} B_{n_i^1}^1 \qquad \text{and} \quad B_{n_i^1+1}^1 \subset \operatorname{Int} B_{n_{i+1}^0}^0.$$

(So the ball of radius $n_i^0 + 1$ for J_0 is contained in the ball of radius n_i^1 for J_1, and the ball of radius $n_i^1 + 1$ for J_1 is contained in the ball of radius n_{i+1}^0 for J_0.) Such subsequences exist because both sequences exhaust M as $n \to \infty$. For each $i \geq 1$, set $V_i^0 := \operatorname{Int} B_{n_i^0+1}^0 \setminus B_{n_i^0}^0$ and $V_i^1 := \operatorname{Int} B_{n_i^1+1}^1 \setminus B_{n_i^1}^1$. These are pairwise disjoint shells escaping to infinity, and each has normalized transverse width at least 1 with respect to the corresponding J_0 or J_1 by (2.1).

We choose a smooth path J_s, $s \in [0,1]$, in $\mathcal{J}_\omega^{\mathrm{comp}}$ from J_0 to J_1 (through ω-compatible structures) which satisfies

$$J_s \equiv J_0 \quad \text{on } \bigcup_i V_i^0 \text{ for } s \in [0, 2/3], \qquad J_s \equiv J_1 \quad \text{on } \bigcup_i V_i^1 \text{ for } s \in [1/3, 1].$$

We claim that each J_s is normalized complete. Indeed, fix $s \in [0,1]$. Then at least one of the two collections of shells, $\{V_i^0\}$ (if $s \leq 2/3$) or $\{V_i^1\}$ (if $s \geq 1/3$), consists of shells on which J_s agrees with a normalized complete structure (either J_0 or J_1) and hence has normalized transverse width ≥ 1. Any d_{norm,J_s}-Cauchy sequence cannot cross infinitely many such shells (each crossing costs at least 1), so it is eventually contained in some $B_{n_i^0}^0$ or $B_{n_i^1}^1$, which is compact in the manifold topology. Since d_{norm,J_s} induces the manifold topology (it is the length metric of a continuous conformal factor $\rho_{J_s}^2 g_{J_s}$ with $\rho_{J_s} > 0$), Cauchy sequences in a compact set converge. Thus $(M, d_{\mathrm{norm},J_s})$ is complete. This shows in particular that $\mathcal{J}_\omega^{\mathrm{nc,comp}}$ is path connected. □

2.3.2. *Normalized confinement and $\boldsymbol{J}$-completeness (see Theorem 0.6 (2)).*

Proof. The proof is an adjustment to Sikorav's confinement argument [14, Proposition 4.4.1], with the bounded-geometry radius replaced pointwise by the scale function ρ_J coming from the local quadratic isoperimetric inequality $\mathcal{I}_J \leq c_0$.

Fix $p \in M$ and set $r := \rho_J(p) \in (0, 1]$. By definition of $r_{\mathrm{iso},J}$ and c_0, every contractible loop $\gamma \subset B_r^{g_J}(p)$ has a smooth filling u with $\int_D u^*\omega \leq c_0\, \ell_{g_J}(\gamma)^2$. The standard monotonicity estimate for J-holomorphic curves (see [14, Proposition 4.3.1]) therefore applies on the ball $B_r^{g_J}(p)$ and gives: if u is a J-holomorphic curve whose image meets p and whose boundary is disjoint from $B_r^{g_J}(p)$, then

$$E\big(u; u^{-1}\big(B_r^{g_J}(p)\big)\big) \geq \frac{r^2}{4c_0}. \tag{2.2}$$

Let $u\colon (\Sigma, \partial\Sigma) \to (M, \omega)$ be a connected J-holomorphic curve, and set $A := u(\partial\Sigma)$. We define the core of the image as the set of points sufficiently far from the boundary:

$$U := \{x \in u(\Sigma) \mid d_{g_J}(x, A) \geq \rho_J(x)\}.$$

For every $x \in U$, the extrinsic ball $B^{g_J}_{\rho_J(x)}(x)$ is disjoint from A. Consequently, applying the monotonicity estimate (2.2) to each such ball guarantees that the energy of the curve within it is at least $\rho_J(x)^2/4c_0$.

Consider the family of extrinsic balls $\mathcal{B} = \big\{B^{g_J}_{\rho_J(x)}(x)\big\}_{x\in U}$. By the Vitali covering lemma, there exists a countable subcollection

$$\big\{B_i = B^{g_J}_{a_i}(x_i)\big\}_{i\in I}, \qquad \text{with } a_i = \rho_J(x_i),$$

consisting of pairwise disjoint balls such that their 5-times expansions $\widehat{B}_i = B^{g_J}_{5a_i}(x_i)$ completely cover U. Because the balls B_i are pairwise disjoint in M, their preimages are disjoint in Σ. Summing the energy over this subcollection avoids any overcounting, yielding

$$\sum_{i\in I} a_i^2 \leq 4c_0 \sum_{i\in I} E\big(u; u^{-1}(B_i)\big) \leq 4c_0 E(u).$$

We now estimate the $d_{\mathrm{norm},J}$-diameter of the expanded balls $\widehat{B}_i$. For any two points $y, z \in \widehat{B}_i$, consider a g_J-geodesic γ connecting them; its length satisfies $L \leq 10a_i$. Since ρ_J is 1-Lipschitz, the value of the scale function along γ is bounded by $\rho_J(x_i) + d_{g_J}(x_i, \gamma(t)) \leq a_i + 5a_i = 6a_i$. The normalized distance is thus bounded by the integral of the scale along γ

$$d_{\mathrm{norm},J}(y, z) \leq \int_\gamma \rho_J(\gamma(t))\mathrm{d}t \leq (6a_i) \cdot L \leq 60a_i^2.$$

Thus, the normalized diameter of each $\widehat{B}_i$ is at most $60a_i^2$. Because the normalized metric integrates the scale, the normalized diameter of $\widehat{B}_i$ is bounded by its maximum scale ($6a_i$) times its g_J-diameter ($10a_i$), which is at most $60a_i^2$.

For any point $p \in u(\Sigma)$, we can now bound its normalized distance to the boundary A. If $p \notin U$, then $d_{g_J}(p, A) < \rho_J(p)$. A g_J-shortest path from p to A has g_J-length at most $\rho_J(p)$, and the scale along this path is at most $2\rho_J(p)$. Thus, its normalized length is at most $2\rho_J(p)^2 \leq 2$. If $p \in U$, any path in M connecting p to A must exit U. We can trace a path to the boundary of U entirely within the cover $\big\{\widehat{B}_i\big\}$. The normalized distance to traverse this cover is strictly bounded by the sum of the normalized diameters of all the expanded balls. Once outside U, the remaining normalized distance to A is at most 2.

We can therefore unconditionally estimate the normalized distance from any $p \in u(\Sigma)$ to the boundary A by combining these bounds

$$d_{\mathrm{norm},J}(p, A) \le \sum_{i \in I} 60a_i^2 + 2 \le 240c_0 E(u) + 2. \qquad \square$$

In the formulation of Theorem 0.6 (2), we only dealt with the larger energy regime. Here we record a localized version which includes also the low-energy regime.

Lemma 2.9 (normalized local confinement)**.** *Let J be an ω-compatible almost complex structure that is normalized complete. There exist universal constants $C_1, C_2 > 0$ such that for every connected J-holomorphic curve $u\colon (\Sigma, \partial\Sigma) \to (M, \omega)$,*

$$u(\Sigma) \subset B_\rho^{d_{\mathrm{norm},J}}(u(\partial\Sigma)), \qquad \rho := \max\bigl(C_1\, E(u), C_2\, \sqrt{E(u)}\bigr).$$

In particular, as $E(u) \to 0$ the $d_{\mathrm{norm},J}$-diameter of $u(\Sigma)$ tends to zero. Moreover, this containment is unaffected by the behavior of J outside the normalized ball of radius ρ.

Proof. We revisit the end of the proof of Theorem 0.6 (2). Using the same notation, the normalized distance from any point $p \in u(\Sigma)$ to the boundary $A = u(\partial\Sigma)$ is at most the sum of the normalized diameters of the expanded balls $\widehat{B}_i$ covering the core U, plus the normalized distance from an exit point $q \notin U$ to A.

As observed there, the normalized diameter of the cover $\bigcup \widehat{B}_i$ is bounded by $\sum_i 60a_i^2$. By monotonicity, $\sum_i a_i^2 \le 4c_0 E(u)$, so the sum of the diameters is at most $240c_0 E(u)$.

For the final segment from $q \notin U$ to A, the g_J-distance is some length $L < \rho_J(q) \le 1$. Because the scale function is globally bounded by 1, the normalized distance of this shortest path is at most its g_J-length L. Furthermore, the standard monotonicity estimate applied to the ball $B_L^{g_J}(q)$ gives $E(u) \ge L^2/4c_0$. Thus, the normalized distance from q to A is bounded by $L \le \sqrt{4c_0 E(u)}$.

Combining these yields

$$d_{\mathrm{norm},J}(p, A) \le 240c_0 E(u) + \sqrt{4c_0 E(u)}.$$

If $E(u) \ge 1$, we have $\sqrt{E(u)} \le E(u)$, and we can absorb the square root term into the linear term by taking $C_1 = 240c_0 + \sqrt{4c_0}$. If $E(u) < 1$, we have $E(u) < \sqrt{E(u)}$, and we can absorb the linear term to get an estimate $\le C_2\sqrt{E(u)}$ by taking $C_2 = 240c_0 + \sqrt{4c_0}$.

The last sentence is immediate since the monotonicity estimate depends only on the local behavior. $\square$

2.3.3. *Equivalence to standard metric under uniform bounds (Theorem 0.6 (3)).*

Proof. Under the assumption, g_J is a-bounded at every point $p \in M$. See Definition 1.3. So the scale function satisfies $\rho_J(p) \in [2/a, 1]$. See [6, Lemma 4.10]. Therefore, the scaled metric is bi-Lipschitz equivalent to the metric g_J. $\square$

Remark 2.10 (ω-tame case)**.** If instead of ω-compatible one uses an ω-tame almost complex structure that is geometrically bounded and satisfies the uniform tameness condition $|\omega|_{g_J} \le C$, then one still obtains an isoperimetric scale and scale function as in Section 2.1, but with a cutoff constant c_0' depending on C in place of $c_0 = \frac{1}{2}$. See [12, Section 4.4] and [7, Section 3, Appendix B]. The statements of Theorem 0.6 (2) and (3) then hold with the same proofs after adjusting constants.

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References

[1] Auroux D., Mirror symmetry and T-duality in the complement of an anticanonical divisor, *J. Gökova Geom. Topol. GGT* **1** (2007), 51–91, arXiv:0706.3207.

[2] Cho C.-H., Oh Y.-G., Floer cohomology and disc instantons of Lagrangian torus fibers in Fano toric manifolds, *Asian J. Math.* **10** (2006), 773–814, arXiv:math.SG/0308225.

[3] Entov M., Ganor Y., Membrez C., Lagrangian isotopies and symplectic function theory, *Comment. Math. Helv.* **93** (2018), 829–882, arXiv:1610.09631.

[4] Fukaya K., Oh Y.-G., Ohta H., Ono K., Lagrangian Floer theory on compact toric manifolds. I, *Duke Math. J.* **151** (2010), 23–174, arXiv:0802.1703.

[5] Gallot S., Hulin D., Lafontaine J., Riemannian geometry, 3rd ed., *Universitext*, Springer, Berlin, 2004.

[6] Groman Y., Floer theory and reduced cohomology on open manifolds, *Geom. Topol.* **27** (2023), 1273–1390, arXiv:1510.04265.

[7] Groman Y., The wrapped Fukaya category for semi-toric Calabi–Yau, *J. Eur. Math. Soc. (JEMS)* **26** (2024), 2373–2439, arXiv:1805.03635.

[8] Groman Y., Varolgunes U., Locality of relative symplectic cohomology for complete embeddings, *Compos. Math.* **159** (2023), 2551–2637, arXiv:2110.08891.

[9] Günther P., Einige Sätze über das Volumen von geodätischen Kugeln, *Ber. Verh. Sächs. Akad. Wiss. Leipzig Math.-Nat. Kl.* **104** (1960), 1–38.

[10] Karshon Y., Tang X., Symplectic excision, arXiv:2101.03534.

[11] McDuff D., Symplectic structures – a new approach to geometry, *Notices Amer. Math. Soc.* **45** (1998), 952–960.

[12] McDuff D., Salamon D., J-holomorphic curves and symplectic topology, 2nd ed., *Amer. Math. Soc. Colloq. Publ.*, Vol. 52, American Mathematical Society, Providence, RI, 2012.

[13] Shelukhin E., Tonkonog D., Vianna R., Geometry of symplectic flux and Lagrangian torus fibrations, *J. Topol.* **17** (2024), e70002, 56 pages, arXiv:1804.02044.

[14] Sikorav J.-C., Some properties of holomorphic curves in almost complex manifolds, in Holomorphic Curves in Symplectic Geometry, *Progr. Math.*, Vol. 117, Birkhäuser, Basel, 1994, 165–189.

[15] Stratmann B., Removing parametrized rays symplectically, *J. Symplectic Geom.* **20** (2022), 499–508, arXiv:2009.05465.